HALO: Zero-Shot Open-Set Human Activity Recognition Across Heterogeneous Sensor Datasets

ABSTRACT

We present HALO (**H**eterogeneity-**A**ware **L**anguage-aligned **O**pen-set model), an IMU foundation model for human activity recognition (HAR) designed to handle sensing heterogeneity and open-set label vocabularies across datasets. Training jointly across multiple HAR datasets is hindered by heterogeneous sampling rates, variable channel counts and placements, and inconsistent label vocabularies. HALO addresses these challenges through three key designs: (i) seconds-based patch segmentation with fixed-length interpolation for handling heterogeneous sampling rates, (ii) channel-independent tokenization with channel-text semantic fusion for accommodating variable sensor configurations, and (iii) CLIP-style contrastive alignment with a frozen Sentence-BERT encoder for zero-shot open-set recognition via text retrieval. Training proceeds in two stages: self-supervised masked patch pretraining followed by IMU-to-text contrastive alignment with synonym-aware soft targets and a memory bank. At inference, activity recognition is cosine similarity retrieval over arbitrary text descriptions—no per-dataset classifier is needed. We train on 10 public HAR datasets and evaluate on 7 held-out datasets under a unified four-metric benchmark spanning zero-shot open-set, zero-shot closed-set, and 1%/10% supervised settings. Against five state-of-the-art baselines, HALO leads on all 8 aggregate metrics across five main test datasets with only ∼35M trainable parameters—10× fewer than the closest competitor MOMENT (341M). Rich channel-text conditioning at native sampling rates contributes 11.4 additional percentage points of zero-shot accuracy on average. We also report two severe out-of-domain datasets where all models collapse in zero-shot, highlighting the limits of cross-dataset transfer under genuine distribution shift.

1 INTRODUCTION

Inertial measurement units (IMUs) are among the most widely deployed sensors in consumer electronics, enabling human activity recognition (HAR) for applications in health monitoring [2], rehabilitation [26], sports analytics [3], and human-computer interaction. Training effective HAR models faces a fundamental heterogeneity problem: publicly available datasets differ in sampling rate (20–100 Hz), sensor suite (3 to 51 channels spanning accelerometers, gyroscopes, and magnetometers at different body placements), and label vocabulary (dataset-specific activity names that overlap inconsistently). A model trained on one dataset rarely transfers well to another, and training jointly across many datasets requires handling conflicting sensor configurations and label inventories simultaneously.

Addressing this challenge requires a model that can (i) operate at arbitrary sampling rates without architectural modifications, (ii) handle variable sensor suites across devices, and (iii) recognize activities absent from training—i.e., support open-vocabulary recognition under label shift. Existing approaches address these axes in isolation. Per-dataset supervised models reach strong intra-dataset performance but do not transfer [21]. Self-supervised pretraining methods such as LiMU-BERT [37] and CrossHAR [4] improve cross-dataset transfer, but still require a closed-world classifier over training labels, leaving them unable to recognize new activities at test time. General time-series foundation models such as MOMENT [8] achieve broad coverage, but are not specifically designed for HAR and equally require downstream classifiers. Language-aligned approaches like LanHAR [14] decouple the label interface using text, but without strong sensor pretraining, text alignment alone proves insufficient for robust zero-shot transfer.

We present HALO, an IMU foundation model for HAR that addresses all three axes simultaneously by treating language as the label interface. HALO is trained jointly across 10 heterogeneous HAR datasets using a CLIP-style [24] contrastive objective that aligns IMU segment embeddings with natural-language activity descriptions in a shared 384-dimensional space. At inference, recognition is

cosine similarity retrieval over a text embedding library, requiring no per-dataset classifier and supporting arbitrary label inventories at test time.

The key idea is to design each component around a specific heterogeneity challenge, so that a single model can train jointly across diverse datasets without any dataset-specific modification. First, channel-independent CNN tokenization with shared weights accommodates variable sensor configurations (3 to 45 channels), while seconds-based patch segmentation with fixed-length interpolation decouples the model from any specific sampling rate. Second, a channel-text semantic fusion module conditions each channel on its sensor description (e.g., "Accelerometer X-axis, waist-mounted smartphone"), injecting placement context via learned gated fusion. Third, CLIP-style contrastive alignment [24] with a frozen SentenceBERT backbone [25] maps IMU segments and text descriptions into a shared space, enabling open-set recognition via cosine similarity retrieval. Synonym-aware soft targets allow semantically related labels ("walking", "strolling", "ambulating") to contribute coherently to training, and a memory bank [10] provides additional hard negatives per step.

Training proceeds in two stages. Stage 1 pretrains the IMU encoder (CNN and temporal Transformer) using masked patch autoencoding and patch-level contrastive learning. Stage 2 trains the full IMU-to-text alignment system initialized from Stage 1.

We evaluate HALO on seven held-out test datasets—five main benchmarks with high label coverage and two severe out-of-domain datasets—under four settings: zero-shot open-set (predicting from all 87 training labels), zero-shot closed-set (predicting from test-dataset labels), and 1% and 10% supervised fine-tuning. Against five state-of-the-art baselines spanning reconstruction-based, text-aligned, and LLM-based paradigms, HALO leads on all 8 aggregate metrics across the five main datasets while using only ∼35M trainable parameters (10× fewer than MOMENT). In zero-shot recognition, HALO leads all baselines; native-rate evaluation with rich channel descriptions adds 11.4 percentage points of zero-shot accuracy. On two severe out-of-domain datasets, all models collapse in zero-shot, but HALO adapts effectively with supervised fine-tuning.

We make the following contributions:

(1) **HALO**: an IMU foundation model for HAR supporting native-rate inference, variable channel counts, and open-vocabulary zero-shot recognition via language alignment.
(2) **Channel-text conditioning**: a module that fuses per-channel semantic descriptions (sensor type and placement) into IMU representations via a learned sigmoid gate, enabling context-aware sensor encoding.
(3) **Unified four-metric benchmark**: a cross-dataset evaluation framework across 10 training and 7 test datasets comparing text-aligned, classifier-based, and LLM-based models across zero-shot and supervised regimes.
(4) **Honest analysis**: we document both gains and limitations, including a negative result for RealWorld HAR and two severe OOD datasets where all models fail in zero-shot, identifying per-sample metadata conditioning and domain adaptation as concrete directions for future work.

2 RELATED WORK

IMU-based HAR.. Deep learning for HAR has progressed from single-dataset CNN-LSTM models [21] to approaches addressing practical heterogeneity: Cosmo [22] demonstrates contrastive fusion for small-data multimodal HAR, UniHAR [36] tackles cross-user and cross-device generalization via physics-informed augmentation, and ClusterFL [23] captures inter-user relationships through federated learning. Despite these advances, all remain architecturally bound to a fixed channel count, sampling rate, and closed-world label set, limiting transferability across sensing setups.

Self-supervised pretraining for wearable sensing. Self-supervised objectives reduce dependence on labeled IMU data. Contrastive methods learn augmentation-invariant representations [5], while masked autoencoding has been adapted from vision [9] and language [6] to IMU sequences: LiMU-BERT [37] pretrains via span-masked reconstruction at 20 Hz, and CrossHAR [4] combines masked reconstruction with contrastive learning for cross-dataset HAR. PatchTST [20] demonstrates that patch-based channel-independent tokenization improves time-series modeling, a design principle we adopt. However, fixed-length positional encodings constrain variable sampling rates, and all methods require a dataset-specific classifier.

Foundation models for time series and tabular data. Foundation models for general time-series data aim for broad coverage across domains. MOMENT [8] is a T5-style encoder pretrained on 13 time-series domains via masked patch reconstruction (∼385M parameters). UniMTS [39] targets unified pretraining for motion time series across heterogeneous sensor configurations. In the tabular domain, Mitra [38] demonstrates that curated mixtures of synthetic priors can train foundation models that generalize across diverse real-world datasets via in-context learning, illustrating the broader trend of domain-agnostic foundation models. These general-purpose approaches trade domain-specific inductive biases for breadth; HALO occupies an intermediate position as a domain-specific foundation model for wearable sensing with HAR-specific designs and a language-based label interface.

Language-grounded perception. CLIP [24] demonstrated that aligning vision and text in a shared embedding space enables open-vocabulary zero-shot classification without per-class classifiers. ImageBind [7] extends this to multiple modalities including IMU, but is not designed for cross-dataset HAR or variable sensor configurations. LanHAR [14] applies CLIP-style alignment to HAR by training a sensor Transformer against LLM-generated activity descriptions. LLaSA [11] takes an alternative approach, leveraging LLMs directly for natural language reasoning about IMU data. HALO combines self-supervised sensor pretraining with CLIP-style alignment, Perceiver-style [12] cross-channel fusion, and architectural mechanisms for native sampling rate handling and channel-text conditioning.

3 MOTIVATION

We identify three key challenges that prevent IMU-based HAR models from generalizing across realistic deployment scenarios, and explain how each one shapes a specific design decision in HALO.

3.1 Sensing Heterogeneity Makes Joint Training Difficult

Publicly available HAR datasets differ along three axes that together make naïve joint training impractical. First, sampling rates vary widely: our ten training datasets span 20 Hz to 100 Hz, meaning that a raw 4-second window contains anywhere from 80 to 400 timesteps depending on the dataset. A fixed positional encoding tied to timestep indices conflates physical time with sequence position, causing the model to see the same motion at 50 Hz and 100 Hz as structurally different inputs. Second, channel counts and sensor placements vary from 3-axis accelerometers on a single wrist to 51-channel multi-IMU setups with gyroscopes, magnetometers, and multiple body placements simultaneously. A fixed-width input layer cannot accommodate this range without padding artifacts or dataset-specific heads. Third, label vocabularies differ in both naming and granularity: "walking," "level walking," "walking on a flat surface," and "normal walk" all denote the same activity across different datasets, while "sport" and "rope skipping" represent very different levels of specificity.

Figure 1 visualizes this heterogeneity: the 14 datasets span a wide range of sampling rates and channel counts, with most training and test sets clustered around 50 Hz / 6 channels but with outliers at 20–100 Hz and 3–45 channels.

These observations motivate two architectural choices in HALO. Seconds-based patch segmentation followed by interpolation to a fixed 64-step representation decouples the encoder from any specific sampling rate, so that a 1-second patch at 20 Hz and at 100 Hz produce identical-length inputs regardless of the original timestep count. Channel-independent CNN tokenization with shared weights processes each sensor axis independently, enabling the model to handle any number of channels without modification.

3.2 Closed-World Classifiers Cannot Transfer to Novel Activities

Existing self-supervised and supervised approaches share a structural limitation: they produce representations that are consumed by a fixed, dataset-specific classifier. This is adequate within a single dataset but fails in two important transfer scenarios. The first is *label shift*: a model trained on ten datasets with a combined 87 unique activity labels is never exposed to the specific label strings used in a held-out test set, even when the underlying motions overlap. The second is *genuinely novel activities*: our test datasets include activities with no semantic analog in the training label set, making correct prediction structurally impossible for any closed-world model.

In our evaluation, we quantify both failure modes explicitly. For classifier-based baselines (LiMU-BERT, MOMENT, CrossHAR), zero-shot evaluation requires

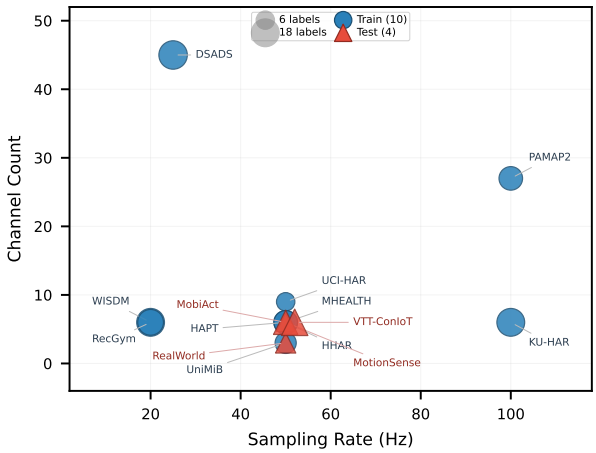

Figure 1: Dataset heterogeneity across the 17 HAR datasets used in this work. Each point represents one dataset; marker size encodes label count. Training datasets (blue circles) and test datasets (red triangles) span 20–100 Hz sampling rates, 3–45 channels, and 4–19 activity labels.

training a linear head over all 87 training labels and then restricting predictions to the subset of labels present in each test dataset; activities in the test set with no training-set counterpart receive a chance floor by construction. Text-aligned models (HALO, LanHAR) avoid this structural limitation entirely: inference is cosine-similarity retrieval over an arbitrary text library, so the candidate set can be changed at runtime without retraining.

3.3 Label Alignment Requires Semantic, Not Syntactic, Matching

Even when the same physical activity appears in both training and test data, the label strings that describe it differ. A model that aligns IMU segments with exact label strings will fail to transfer across datasets that use different phrasing for the same motion. Standard one-hot supervision treats "jogging" and "running" as maximally different, producing contradictory gradients when both labels appear in the training set for similar motions.

HALO addresses this through two complementary mechanisms. At evaluation time, predictions are scored using synonym group membership rather than exact string equality, so semantically correct predictions are not penalized for cross-dataset vocabulary variation. During training, we use cosine similarity between SentenceBERT label embeddings to construct soft target distributions over in-batch candidates rather than hard one-hot assignments. This allows synonymous labels that appear as distinct strings within the same batch to contribute coherently to the contrastive objective rather than generating contradictory gradients.

4 SYSTEM OVERVIEW

HALO is an IMU-to-text semantic alignment framework trained across heterogeneous HAR datasets for open-set, zero-shot activity recognition. Given a batch of IMU sessions, we segment each session into fixed-duration patches and resample each patch to a fixed length. The IMU encoder maps the resulting patch tensor to an embedding $\mathbf{z}_{\text{imu}} \in \mathbb{R}^D$, while a frozen text backbone with learned pooling encodes activity descriptions into $\mathbf{z}_{\text{text}} \in \mathbb{R}^D$. Training optimizes a contrastive objective that increases similarity between matched IMU–text pairs and decreases similarity to non-matching candidates. At inference, recognition is cosine similarity retrieval: given $\mathbf{z}_{\text{imu}}$, we select the label embedding with maximum similarity, supporting arbitrary candidate inventories including open-vocabulary settings. The core technical challenge—heterogeneity in sampling rate, channel configuration, and label vocabulary—is addressed by dedicated mechanisms: seconds-based patching with interpolation, channel-independent tokenization, and language alignment respectively (Sections 3 and 5).

Training proceeds in two stages: Stage 1 pretrains the IMU encoder (CNN + temporal Transformer) via self-supervised patch objectives, and Stage 2 performs semantic alignment with a contrastive objective and memory bank. Figure 2 illustrates the key designs. The full model contains ∼58M parameters (∼35M trainable: 8-layer dual-branch IMU encoder ∼19M, alignment and projection heads ∼15M; plus 22.7M frozen SBERT backbone). For zero-shot evaluation on unseen datasets, we retrieve over a combined label inventory that includes both training labels and test-dataset labels, reflecting deployment where the candidate label space is not fixed.

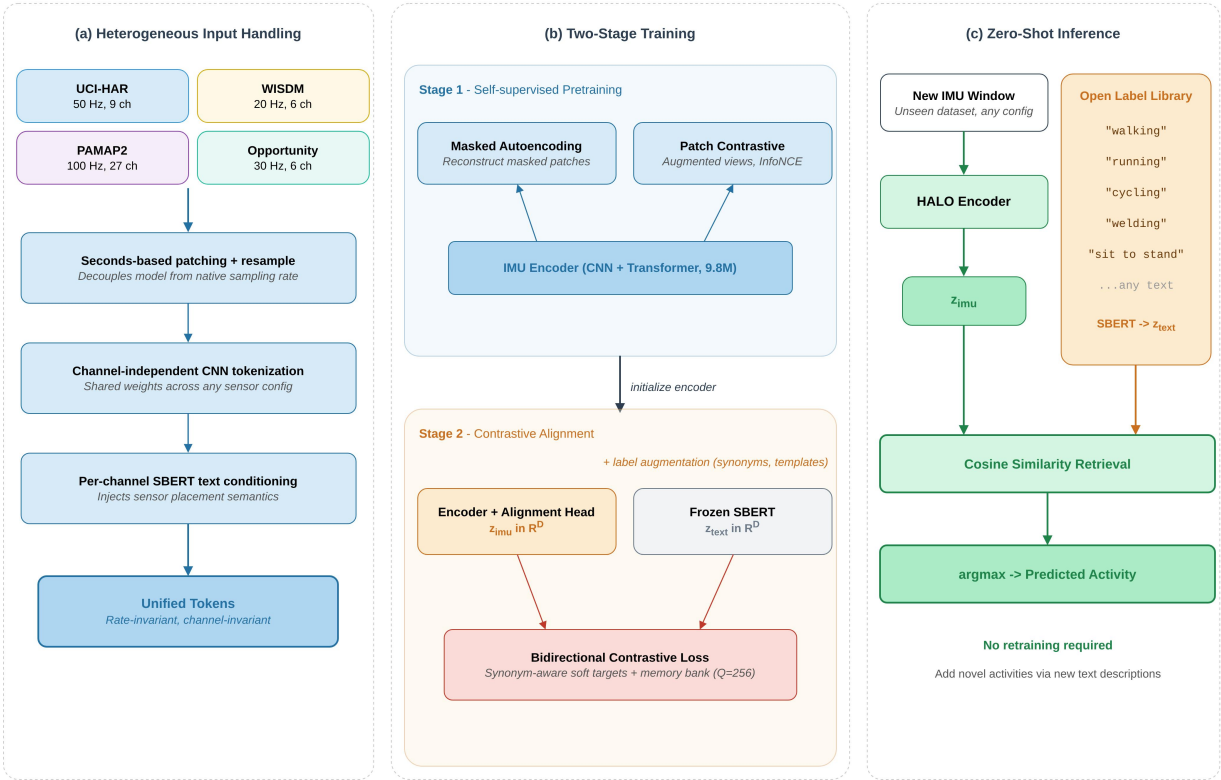

Figure 2: Key designs of the HALO framework. (a) Heterogeneous input handling: seconds-based patching, channel-independent CNN tokenization, and per-channel text conditioning produce rate- and channel-invariant tokens from any IMU configuration. (b) Two-stage training: self-supervised pretraining (Stage 1, blue) initializes the IMU encoder, then contrastive IMU-to-text alignment (Stage 2, orange) with synonym-aware soft targets and a memory bank learns a shared embedding space. (c) Zero-shot inference: cosine similarity retrieval over an open label library enables recognition of novel activities without retraining.

5 DESIGN OF HALO

5.1 Input Representation and Heterogeneity Handling

Notation. A padded batch of sessions has shape $(B, T_{\max}, C_{\max})$. Each session is segmented into P patches of dataset-specific duration and resampled to a fixed length $L = 64$, yielding $\mathbf{X} \in \mathbb{R}^{B \times P \times L \times C}$ (with $C \leq C_{\max}$). The IMU encoder produces a pooled embedding $\mathbf{z}_{\text{imu}} \in \mathbb{R}^{B \times D}$ and the text encoder produces $\mathbf{z}_{\text{text}} \in \mathbb{R}^{B \times D}$, with $D = 384$.

Patching and resampling. HAR datasets differ in sampling rate and session duration. We segment each session into fixed-duration patches (dataset-specific, typically 0.75–2.0s) and resample each patch to $L = 64$ timesteps using linear interpolation. This yields a uniform temporal representation that supports joint training across datasets.

Patch size augmentation (training only). To improve robustness to temporal scale variation, during training we randomly sample patch durations from a dataset-specific discrete range (e.g., 0.75–1.25s for UCI-HAR; 1.0–1.75s for MHEALTH). Augmentation is disabled for validation and evaluation, where each dataset uses a fixed patch size.

Per-patch, per-channel normalization. We apply z-score normalization independently per patch and

per channel:

$$\hat{\mathbf{X}}_{b,p,:,c} = \frac{\mathbf{X}_{b,p,:,c} - \mu_{b,p,c}}{\sigma_{b,p,c} + \epsilon},$$

where $\mu_{b,p,c}$ and $\sigma_{b,p,c}$ are computed over the L timesteps of patch (b, p) channel c.

Variable channels via channel-independent tokenization. Channel counts and sensor suites vary across datasets. We use channel-independent feature extraction with shared parameters across channels. Channels are padded within a batch and masks are propagated to fusion and pooling modules so padded channels do not contribute.

5.2 Model Architecture

Channel-independent CNN tokenizer. Each patch-channel pair is converted into a D-dimensional token via a CNN with shared weights across channels. We reshape $\hat{\mathbf{X}} \in \mathbb{R}^{B\times P\times L\times C}$ into $\mathbb{R}^{(BPC)\times 1\times L}$, apply the CNN stack, and reshape back to obtain tokens $\mathbf{H}_0 \in \mathbb{R}^{B\times P\times C\times D}$. This decouples the tokenizer from dataset-specific channel sets.

Patch positional encoding. We inject temporal order information over patches using sinusoidal positional encodings [34] added to the patch dimension and broadcast across channels. A learnable scale parameter (initialized to a small value) modulates the positional contribution.

Channel-independent temporal Transformer. We model temporal dependencies across patches using a Transformer encoder applied independently per channel. We reshape $\mathbf{H}_0$ from (B, P, C, D) to (BC, P, D), apply L_e layers of bidirectional self-attention over the patch axis, and reshape back to obtain encoded representations $\mathbf{H} \in \mathbb{R}^{B\times P\times C\times D}$.

Channel-text conditioning. Before cross-channel aggregation, each channel token is conditioned on a per-channel semantic description (e.g., "Accelerometer X-axis, waist-mounted smartphone"). The description is encoded by the frozen SentenceBERT backbone into a vector $\mathbf{e}_c \in \mathbb{R}^D$. A learned sigmoid gate computed from the concatenation $[\mathbf{H}_{b,p,c}; \mathbf{e}_c]$ forms the conditioned token as a gated sum of the channel token and description embedding. This injects sensor placement context into each channel's representation before cross-channel fusion.

Cross-channel fusion and temporal pooling. For each patch, we fuse information across channels using multi-query attention: a small set of learnable queries attends over channel tokens, producing a fused patch sequence $\mathbf{U} \in \mathbb{R}^{B\times P\times D}$. Channel masks prevent padded channels from influencing attention. A lightweight Transformer over $\mathbf{U}$ followed by a second multi-query attention module pools the patch axis to yield $\mathbf{v} \in \mathbb{R}^{B\times D}$, which an MLP projection head maps to $\mathbf{z}_{\text{imu}} \in \mathbb{R}^{B\times D}$ (ℓ_2-normalized).

Text encoder. We encode activity descriptions using a frozen Sentence-BERT backbone [25] (all-MiniLM-L6-v2). Learnable query-based pooling (shared queries attending over token sequence outputs) produces $\mathbf{z}_{\text{text}} \in \mathbb{R}^{B\times D}$ (ℓ_2-normalized). Freezing the backbone provides stable semantics and reduces training cost.

5.3 Training Procedure

Stage 1: Self-supervised tokenizer pretraining. Stage 1 trains the IMU encoder before semantic alignment using a dual objective. *Masked autoencoding*: a fixed fraction of patches (mask ratio 0.5) are replaced by a learned mask token and the temporal encoder reconstructs the normalized input signal at masked positions:

$$\mathcal{L}_{\text{mae}} = \frac{1}{|\mathcal{M}|} \sum_{(b,p)\in\mathcal{M}} \|\hat{\mathbf{u}}_{b,p} - \mathbf{u}_{b,p}\|_2^2 .$$

Patch-level contrastive learning: two augmented views of each patch (jittering, scaling, time warping) are treated as positives and other in-batch patches as negatives under an InfoNCE objective [33] with temperature 0.2. Both objectives are combined with equal weights ($\lambda_{\text{mae}} = \lambda_{\text{con}} = 1.0$).

Label augmentation. During Stage 2 training, we augment activity labels with dataset-specific synonyms and natural-language templates to increase linguistic diversity. Each dataset defines a curated synonym map (e.g., *walking* ↔ *strolling* ↔ *ambulating*) and templates (e.g., "person {}"). With probability 0.8 the original label is replaced by a sampled synonym and template; augmentation is disabled at evaluation. This prevents synonymous labels appearing as hard negatives in the contrastive batch, which would generate contradictory gradients.

Stage 2: IMU-to-text contrastive alignment. Stage 2 aligns $\mathbf{z}_{\text{imu}}$ with $\mathbf{z}_{\text{text}}$ via a CLIP-style bidirectional contrastive objective. We maintain FIFO queues [10]

of size $Q = 256$ for both modalities to increase candidates without increasing GPU memory, prefilling them with a warmup pass before training. Similarity logits are computed as

$$\mathbf{S}_{i\to t} = \alpha\, \mathbf{Z}_{\text{imu}}\, \mathbf{A}_{\text{text}}^{\top}, \quad \mathbf{S}_{t\to i} = \alpha\, \mathbf{Z}_{\text{text}}\, \mathbf{A}_{\text{imu}}^{\top},$$

where $\mathbf{A}$ denotes the concatenation of the current batch and queue, and $\alpha = \exp(\texttt{logit_scale})$ is a learnable temperature (CLIP-initialized, clamped to $[1, 50]$).

Adaptive soft targets. Because SentenceBERT similarities between activity phrases fall in a narrow range (0.4–0.9), naive soft targets yield weak discrimination. We z-score-normalize the in-batch text–text similarity matrix $\mathbf{M} = \mathbf{Z}_{\text{text}}\mathbf{Z}_{\text{text}}^{\top}$ with temperature $\tau_s = 0.5$, then softmax to obtain target distributions $\mathbf{T}$. Hard and soft targets are blended ($w = 1.0$ pure soft in all experiments) for in-batch positions; queue entries serve as hard negatives. The bidirectional loss is:

$$\mathcal{L}_{i\to t} = -\frac{1}{B}\sum_{b,j} \mathbf{P}_{b,j} \log \operatorname{softmax}(\mathbf{S}_{i\to t})_{b,j},$$

$$\mathcal{L} = \tfrac{1}{2}\big(\mathcal{L}_{i\to t} + \mathcal{L}_{t\to i}\big).$$

6 EXPERIMENTS

6.1 Methodology

We evaluate whether language-aligned sensor representations support (i) robust performance under heterogeneous multi-dataset training, (ii) open-set and closed-set zero-shot generalization to unseen datasets, and (iii) sample-efficient supervised transfer. We compare against five state-of-the-art baselines spanning reconstruction-based, general-purpose, text-aligned, and generative architectures.

Datasets. We train jointly on ten public HAR datasets spanning diverse sampling rates, channel counts, and label vocabularies (Table 1). We evaluate on seven held-out datasets that are *never seen during any model's training*: five main test datasets with high label-group coverage (85–100%) and two severe out-of-domain datasets (HARTH, sensor distribution shift; VTT-ConIoT, 50% label coverage) reported separately. Figure 3 visualizes this gap: a UMAP projection of raw sensor features shows that every test dataset lies outside the training distribution, with Opportunity occupying an entirely isolated region. HALO evaluates at each dataset's native sampling rate (50 Hz for most; 30 Hz for Opportunity), while baselines receive data resampled to 20 Hz in a standardized $(N, 120, 6)$ format—see the sampling rate policy in Section 6.1.

Dataset	Hz	#Subj	#Act	#Ch	Description
UCI-HAR [1]	50	30	6	9	Smartphone IMU, basic activities
HHAR [31]	∼50	9	6	6	Smartphone + smartwatch IMU
UniMiB-SHAR [19]	50	30	17	3	Smartphone accelerometer, ADLs
MHEALTH [2]	50	10	12	6+	Multi-sensor body IMU suite
PAMAP2 [26]	100	9	18	27	Multi-IMU physical activities
WISDM [13]	20	51	18	6	Smartphone + watch IMU
DSADS [3]	25	8	19	45	Daily/sports, 5 body positions
HAPT [27]	50	30	12	6	Smartphone IMU with transitions
KU-HAR [30]	100	90	18	6	Smartphone IMU, 18 activities
RecGym [40]	20	10	12	6	Smartwatch gym exercises
MotionSense [17]†	50	24	6	6	Smartphone, basic activities
RealWorld [32]†	50	15	8	3	Smartphone/watch, acc-only
MobiAct [35]†	50	66	13	6	Falls and ADLs (novel activities)
Shoaib [29]†	50	10	7	6	Multi-placement smartphone
Opportunity [28]†	30	4	4	6	XSens body-worn IMU
HARTH [15]‡	50	22	12	6	Back-mounted accelerometer
VTT-ConIoT [16]‡	52	12	16	6	Construction worker IMU

Table 1: Dataset overview. Top: 10 training datasets. Bottom: 7 held-out test datasets. † Main test datasets (85–100% label-group coverage). ‡ Severe out-of-domain; reported separately. HALO evaluates at native sampling rates (50 Hz or 30 Hz); baselines use 20 Hz resampled data.

Baselines. We compare HALO against five baselines representing different pretraining paradigms (Table 3).

The baselines fall into three categories, each with different prediction capabilities that constrain what evaluation is meaningful. **Text-aligned models** (HALO, LanHAR) encode activity labels as natural-language strings and predict via cosine similarity between sensor and text embeddings; they can score any arbitrary label string at test time without retraining, but are evaluated by exact string match, which is the strictest criterion. **Classifier-based models**

Test Dataset	Windows	Classes	Group Cov.	Difficulty
MotionSense	12 080	6	100%	Easy
RealWorld	27 138	8	100%	Medium
MobiAct	4 345	13	85%	Hard
Shoaib	5 537	7	100%	Medium
Opportunity	6 453	4	100%	Medium
HARTH	47 330	12	100%	Severe
VTT-ConIoT	2 058	16	50%	Severe

Table 2: Test dataset characteristics. *Group coverage* is the fraction of test activities that have a semantically equivalent group in the 10 training datasets. HARTH achieves 100% coverage but suffers severe sensor distribution shift (back-mounted accelerometer only, no gyroscope). VTT-ConIoT has 8 of 16 novel industrial activities.

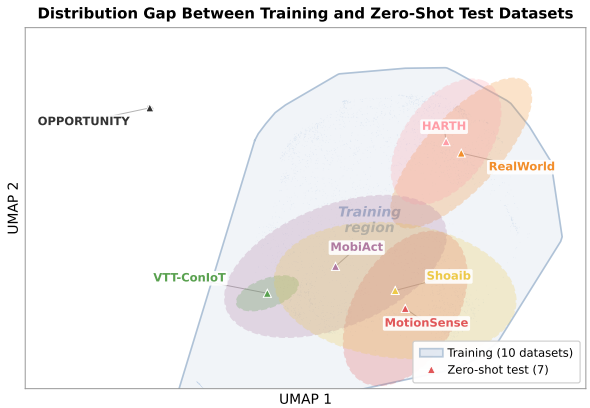

Figure 3: UMAP projection of per-dataset feature centroids computed from raw 6-axis IMU windows across all 17 datasets; ellipses show per-dataset spread. The training distribution (blue hull) is concentrated, while zero-shot test datasets (triangles) scatter across distant regions—particularly Opportunity, which is isolated due to its XSens body-worn configuration. No test dataset overlaps comfortably with the training region, requiring generalization across sensor hardware, placement, and label vocabulary.

(LiMU-BERT, MOMENT, CrossHAR) train a fixed-output classifier over the 87 unique global labels from the 10 training datasets; they cannot predict unseen label strings, but benefit from group-based scoring that credits semantically equivalent predictions (e.g., predicting "jogging" when the ground truth is "running"). **Generative models** (LLaSA) prompt a 7B LLM with sensor tokens and an activity list, then parse the generated text; responses that cannot be matched to any valid label are scored as incorrect. These asymmetries are inherent to the architectures, not artifacts of our evaluation design, and are the central motivation for text-aligned sensing.

Zero-shot classifier training (classifier-based models only). For zero-shot evaluation of LiMU-BERT, MOMENT, and CrossHAR, a classifier over the 87 training labels is trained on embeddings extracted from the 10 training datasets—no test data is used at any point. LiMU-BERT uses a 2-layer GRU head (100 epochs, 90/10 split by session); MOMENT uses an SVM-RBF classifier trained with 5-fold cross-validation on frozen embeddings; CrossHAR uses its paper's Transformer_ft classifier head (100 epochs, 90/10 split). The term *zero-shot* refers to the test dataset being unseen at both training and classifier-fitting stages, not to the classifier being untrained—analogous to how CLIP trains on paired image–text data and then evaluates zero-shot on ImageNet.

Training data. All HAR-pretrained models (HALO, LiMU-BERT, CrossHAR, LanHAR) are trained on the same 10 training datasets. MOMENT is pre-trained on general time-series data with no HAR-specific data. LLaSA uses published weights trained by its authors on HHAR, MotionSense, Shoaib, and UCI-HAR; because MotionSense and Shoaib overlap with our test datasets, LLaSA results on those two are marked with * to indicate they are not truly zero-shot. All seven test datasets are strictly held out from every other model's training.

Evaluation protocol. We use a unified four-metric evaluation framework designed to measure different aspects of representation quality.

Group-aware scoring and unmappable activities. HAR datasets use inconsistent label strings for semantically equivalent activities (e.g., "jogging" in one dataset, "running" in another; "climbing stairs" vs. "walking upstairs"). We define *synonym groups* that map variant strings to a canonical activity (e.g., group *running* = {*running, jogging, nordic_walking*}). Classifier-based models can only output training labels, so without grouping they would be penalized for vocabulary mismatch even when their prediction is semantically correct (e.g., predicting "jogging" for

Model	Dim	Trainable	Inference	Approach
LiMU-BERT [37]	72	72.7K	72.7K	BERT-style span-masked reconstruction on 120-step (6 s) 20 Hz IMU windows; GRU downstream classifier (see Table 4).
MOMENT [8]	6144	341.2M	341.2M	T5-style encoder pretrained on 13 diverse time-series domains; processes each IMU channel independently and concatenates $6 \times 1024 = 6144$-dim (no fusion).
CrossHAR [4]	72	531.4K	531.4K	Hierarchical self-supervised pretraining combining span-masked reconstruction and patch-level contrastive regularization for cross-dataset HAR.
LanHAR [14]	768	13.0M*	122.9M	Two-stage CLIP-style alignment: fine-tune SciBERT on activity text, then train a sensor Transformer to align with text representations.
LLaSA [11]	—	∼6.7B	∼6.7B	LIMU-BERT IMU encoder + Vicuna-7B LLM; classifies by prompting the LLM with sensor tokens and parsing generated text. Zero-shot only.
HALO (ours)	384	∼35M	∼58M	8-layer dual-branch Transformer with per-patch prediction and majority voting; CLIP-style contrastive alignment with frozen SentenceBERT (22.7M), synonym-aware soft targets, and a memory bank; evaluated at native sampling rates.

Table 3: Baseline overview. *Dim*: embedding dimensionality. *Trainable*: parameters updated during pretraining/alignment. *Inference*: total parameters loaded at test time including frozen modules. HALO's inference total includes a frozen 22.7M SentenceBERT encoder. *LanHAR freezes SciBERT (109.9M) during stage-2 training; unfreezes for supervised fine-tuning. All HAR-pretrained models use the same 10 training datasets; MOMENT uses general time-series data; LLaSA uses published weights trained on HHAR, MotionSense, Shoaib, and UCI-HAR.

a test sample labeled "running"). Group-based scoring removes this penalty and is therefore a fairness concession *in favour of* classifier-based models. Text-aligned models in closed-set evaluation are scored by exact label match over the test dataset's own vocabulary—a strictly harder criterion, since a near-miss retrieval (e.g., returning "jogging" instead of "running") counts as wrong.

Some test activities have *no* semantic equivalent in any training group. For these *novel activities*, accurate prediction is structurally impossible for all models: classifier-based models cannot output any label that maps to them, and text-aligned models have no meaningful training signal to associate their sensor patterns. Coverage of the five main test datasets is high (MotionSense, RealWorld, Shoaib, Opportunity: 100%; MobiAct: 85%—two novel activities: *car_step_in*, *car_step_out*). The two severe OOD datasets present different challenges: HARTH has 100% label coverage but severe sensor distribution shift (back-mounted accelerometer only, no gyroscope, gravity-contaminated), and VTT-ConIoT has 50% novel industrial activities (e.g., *kneeling work*, *roll painting*). These conditions form irreducible error floors; both datasets are reported separately.

Zero-shot open-set. Each model predicts from the full 87-label training vocabulary. *Text-aligned:* cosine similarity over all 87 text embeddings → argmax → group-level match to ground truth. *Classifier-based:* native classifier 87-class logits → argmax → group-level match. Novel test activities are guaranteed wrong under open-set for classifier-based models; text-aligned models can attempt retrieval but have no in-vocabulary match available.

Zero-shot closed-set. Each model's prediction space is *restricted to labels relevant to the test dataset*, simulating a scenario where the candidate set is known. *Text-aligned:* encode only the test dataset's own label strings → cosine similarity → argmax → exact match. Because the test labels are used directly as text, novel activities are included in the candidate set for text-aligned models (though performance on them remains low). *Classifier-based:* the 87-class logit vector is *masked* to zero out all positions except training labels whose synonym group is represented in

the test dataset. Prediction proceeds by argmax over this masked vector → group-level match. For MotionSense, 22 of 87 training labels are permitted; for RealWorld, 30; for MobiAct, 34; for Shoaib, 25; for Opportunity, 14; for HARTH, 29. Crucially, the mask cannot include novel activities (MobiAct's two car-entry activities, all 8 VTT-ConIoT industrial activities): no training label maps to their groups, so they remain guaranteed errors for classifier-based models regardless of model quality. This closed-set masking makes the prediction task easier (fewer confusable labels) while remaining honest about what classifier-based models can and cannot predict.

1% supervised. The encoder is fine-tuned end-to-end using 1% of labeled data from the test dataset (balanced subsampling per class, at least one sample per class, 80/10/10 window-level train/val/test split, seed 3431). All models use the same training loop: 20 epochs, AdamW, encoder learning rate 10^{-5}, head learning rate 10^{-3}, batch size 32, early stopping with patience 5 on validation accuracy. *Text-aligned models* (HALO, LanHAR) fine-tune the sensor encoder and classify by cosine similarity with frozen text embeddings—no separate classifier head is trained. *LiMU-BERT* fine-tunes the encoder jointly with its 2-layer GRU head. *MOMENT* fine-tunes the encoder jointly with a linear projection head. *CrossHAR* fine-tunes the encoder jointly with its Transformer_ft classifier.

10% supervised. Identical to 1% supervised with 10% labeled data, testing the semi-supervised regime.

Sampling rate policy. Our protocol evaluates each model at its architectural best: every model receives the richest input its architecture can consume. HALO's seconds-based patch tokenization with interpolation to a fixed 64-step representation decouples it from any specific rate, so it evaluates at native rates (50 Hz for most datasets, 30 Hz for Opportunity) with per-dataset channel descriptions derived from dataset manifests (e.g., "Accelerometer X-axis (waist-mounted smartphone)"). LiMU-BERT, CrossHAR, and LLaSA use learned positional embeddings fixed at 120 positions for 20 Hz and cannot accept higher rates without retraining. MOMENT processes raw number sequences with no concept of physical time. LanHAR trains its sensor encoder from scratch and adapts to whatever rate it receives. These baselines therefore evaluate on 20 Hz resampled data—not because we withhold data, but because their architectures cannot use it. Conversely, MOMENT benefits from its 341M parameters and 6144-dim embeddings, and classifier-based models benefit from group-based scoring; each of these is an architectural advantage we preserve rather than neutralize.

Fairness summary. Several design choices jointly ensure that no model is systematically advantaged or disadvantaged. (i) All models evaluate on identical windows, labels, and data splits (seed 3431); no model sees test data during training. (ii) Each baseline uses its own paper's recommended downstream classifier so that transfer quality is measured through each model's intended path. (iii) Wherever we deviated from a baseline's original protocol, we erred in that baseline's favour (e.g., unfreezing CrossHAR's encoder, giving MOMENT end-to-end fine-tuning; see Table 4). (iv) Group-based scoring benefits classifier-based models; exact-match scoring for text-aligned models is strictly harder. (v) Each model receives the richest input its architecture can consume (native rates for HALO, 6144-dim embeddings for MOMENT, group scoring for classifier-based models). We report both accuracy and macro-averaged F1 for every metric so that class-imbalance effects are visible.

Adaptations from original papers. We adapt each baseline to our unified benchmark rather than replicating each paper's exact protocol. Table 4 documents every significant deviation with a fairness rationale. The overarching principle is that a cross-baseline benchmark requires standardized evaluation units, identical data splits, and comparable fine-tuning conditions; replicating each paper's exact protocol would produce numbers on different datasets, different splits, and different evaluation granularities.

Training protocol. Training proceeds in two stages with a single model trained jointly across all 10 datasets.

Stage 1 tokenizer pretraining. We pretrain the IMU encoder using the dual objective described in Section 5.3. The default configuration uses batch size 20, 100 epochs, AdamW with learning rate 10^{-4} and weight decay 10^{-5}, and 10 warmup epochs. The MAE mask ratio is 0.5 and the patch-level contrastive temperature is 0.2.

Model	What changed	Original paper	Our adaptation	Net effect
LiMU-BERT	Window scoring	Each 20-step sub-window scored by the GRU classifier independently; final label is per-window	Majority vote across the 6 non-overlapping sub-windows that make up one 120-step evaluation window	Neutral (standardizes evaluation unit across models)
LiMU-BERT	Model scope	Separate per-dataset pretrained models; each model trained on one dataset	Single model pretrained jointly on all 10 training datasets	Neutral; enables fair cross-dataset comparison
CrossHAR	Supervised fine-tuning	Encoder frozen during fine-tuning; only the lightweight Transformer_ft classifier head is trained	Full encoder + head fine-tuned end-to-end with a lower encoder learning rate (10^{-5})	Slight advantage to CrossHAR (encoder adapts to target distribution)
MOMENT	Supervised method	SVM-RBF trained on frozen MOMENT embeddings; encoder not fine-tuned for classification tasks	Linear projection head fine-tuned end-to-end with the encoder under AdamW	Slight advantage to MOMENT (encoder adapts to target)
LanHAR	Training data scope	Sensor encoder trained on source *and* target domain data (cross-domain adaptation setup)	Source data only; all seven test datasets strictly excluded from training and fine-tuning	Slight disadvantage to LanHAR; necessary to prevent test-set data leakage
LanHAR	Supervised protocol	Original paper evaluates zero-shot generalization only; no supervised fine-tuning reported	Cosine-similarity fine-tuning added under the same protocol as HALO for completeness	Extension; results are additional, not a replication claim
LLaSA	Published weights	Paper trains on HHAR, MotionSense, Shoaib, UCI-HAR with GPT-generated narrations	Published BASH-Lab/LLaSA-7B (HuggingFace, fp16); no retraining	Retraining 7B LLM infeasible for baseline comparison
LLaSA	Zero-shot only	Paper evaluates zero-shot classification	Zero-shot open-set and closed-set; no supervised fine-tuning	Fine-tuning 7B LLM with 1–10% labels is impractical
LLaSA	Subsampled eval	Paper uses 10 samples per class	Up to 100 samples per class (vs. full dataset for other baselines)	LLM inference at ∼0.3 s/sample; 100/class is 10× the paper's protocol

Table 4: Deviations from each baseline's original evaluation protocol, with a fairness rationale for each. Adaptations that advantage a baseline (CrossHAR, MOMENT) err toward those baselines. The LanHAR data restriction is necessary to prevent test-set leakage and does not reflect the paper's intended protocol.

Stage 2 semantic alignment. We initialize Stage 2 from the pretrained encoder unless otherwise stated. We use a 70/15/15 train/val/test split for each training dataset and train a single model jointly. Each

micro-step draws a batch of sessions that are patched and resampled to 64 timesteps. Training uses mixed precision and gradient accumulation (micro-batch 32, accumulation 16).

Class balancing. We use group-balanced sampling to mitigate class imbalance across datasets. Activity labels are mapped to semantic groups and sampling weights are set inversely proportional to group frequency, with oversampling capped at 20×.

Key hyperparameters. Default values: learned temperature α_0=0.07 (CLIP-style init), soft-target temperature τ_s=0.5, soft-target weight w=1.0, memory bank Q=256, gradient accumulation over 16 micro-steps.

Evaluation patch size. HALO uses a fixed 1.0 s patch size for all test datasets—a metadata-only decision with no per-dataset tuning. At native rates (50 Hz for most datasets, 30 Hz for Opportunity), this yields 30–50 timestep patches interpolated to 64 fixed steps, providing fine temporal resolution while producing 6 tokens per channel. Sensitivity to this choice is analyzed in Section 6.4.

6.2 Overall Performance

6.2.1 Average Performance Across Main Datasets. Table 5 summarizes average performance across the five main test datasets.

HALO leads on all eight metrics in the average table. In zero-shot open-set retrieval over 87 candidate labels, HALO reaches 46.0% accuracy, ahead of MOMENT (28.3%) by nearly 18 percentage points. In the closed-set setting, HALO leads both accuracy (53.4% vs. MOMENT's 44.7%) and F1 (39.4 vs. 33.2), a margin of 8.7 and 6.2 points respectively. At 1% supervised fine-tuning, HALO leads accuracy (77.5% vs. MOMENT's 73.8%) and F1 (71.4 vs. 69.8), despite MOMENT's 6144-dim embeddings (16× HALO's 384-dim) giving its linear head substantially more capacity. At 10%, HALO leads in both accuracy (85.3% vs. 81.3%) and F1 (81.8 vs. 76.8).

CrossHAR is a competitive third despite its compact architecture (531K parameters, 72-dim embeddings), achieving 78.6% at 10% supervision. LiMU-BERT struggles severely at 1% (35.7%), suggesting its GRU classifier needs more labeled data to converge; performance remains limited at 10% (51.3%). LanHAR underperforms across all metrics despite being text-aligned, likely because its BERT text encoder is fine-tuned from scratch on small HAR data and its sensor encoder lacks self-supervised pretraining. LLaSA performs worst overall (17.4% closed-set average), confirming that generative LLM-based classification underperforms direct embedding comparison for IMU data.

6.2.2 Per-Dataset Results. Figure 4 reports per-dataset accuracy across all four evaluation settings for the five main test datasets.

Original benchmarks (MotionSense, RealWorld, MobiAct). HALO leads zero-shot on all three original datasets. On MobiAct—the hardest original dataset with 13 classes including falls and vehicle entry—HALO leads both open-set (49.8%) and closed-set (59.3%), ahead of MOMENT by 21.1 and 18.4 percentage points respectively. On MotionSense, HALO's open-set accuracy of 57.4% leads MOMENT's 33.8% by a wide margin. On RealWorld, HALO now leads both open-set (49.8%) and closed-set (46.3%), ahead of CrossHAR (40.3%), a reversal from the earlier model where dataset-level channel descriptions hurt performance on this heterogeneous multi-placement dataset (see Section 6.3).

Expanded benchmarks. On Shoaib (7 activities, multi-placement), HALO leads across all metrics, including open-set (46.0%), closed-set (48.0% vs. MOMENT's 46.0%), 1% supervised (89.0% vs. 87.6%), and 10% (95.3% vs. 92.8%). Opportunity (XSens body-worn IMUs at 30 Hz, 4 basic activities) is the only main dataset where MOMENT leads HALO in zero-shot closed-set (53.9% vs. 50.4%); MOMENT's 6144-dim SVM-RBF classifier with group-based scoring provides a strong edge on this small, well-separated label set. HALO leads at 10% supervision on Opportunity (76.5% vs. 73.4%).

Supervised results across all five datasets. At 10% supervision, HALO leads on all 5 datasets. At 1%, HALO leads on 4 of 5 datasets; MOMENT leads only on Opportunity (74.3% vs. 71.8%), where its high-dimensional embeddings provide an edge on this small, well-separated label set.

6.2.3 Severe Out-of-Domain Transfer. Table 6 reports HARTH and VTT-ConIoT separately because they test conditions fundamentally different from the main benchmarks. HARTH uses back-mounted and thigh-mounted accelerometers only (no gyroscope), with gravity-contaminated raw data that differs substantially from the waist/wrist-mounted smartphone IMUs in training data; despite 100%

Model	ZS Open-Set		ZS Closed-Set		1% Supervised		10% Supervised	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
HALO (ours)	**46.0**	**16.9**	**53.4**	**39.4**	**77.5**	**71.4**	**85.3**	**81.8**
MOMENT	28.3	7.2	44.7	33.2	73.8	69.8	81.3	76.8
CrossHAR	21.2	5.5	38.2	31.7	61.1	55.2	78.6	75.2
LiMU-BERT	21.7	7.5	33.1	23.1	35.7	22.7	51.3	43.6
LanHAR	15.8	6.2	28.4	21.7	43.3	35.2	56.2	51.5
LLaSA†	1.4	1.2	17.4	8.4	—	—	—	—

Table 5: Average accuracy (%) and macro F1 (%) across 5 main test datasets (MotionSense, RealWorld, MobiAct, Shoaib, Opportunity). Bold: best per metric. HALO evaluates at native rates with rich channel descriptions; baselines use 20 Hz resampled data. †LLaSA: published 7B model, zero-shot only (no supervised), subsampled to 100 samples/class; MotionSense and Shoaib are in LLaSA's training data.

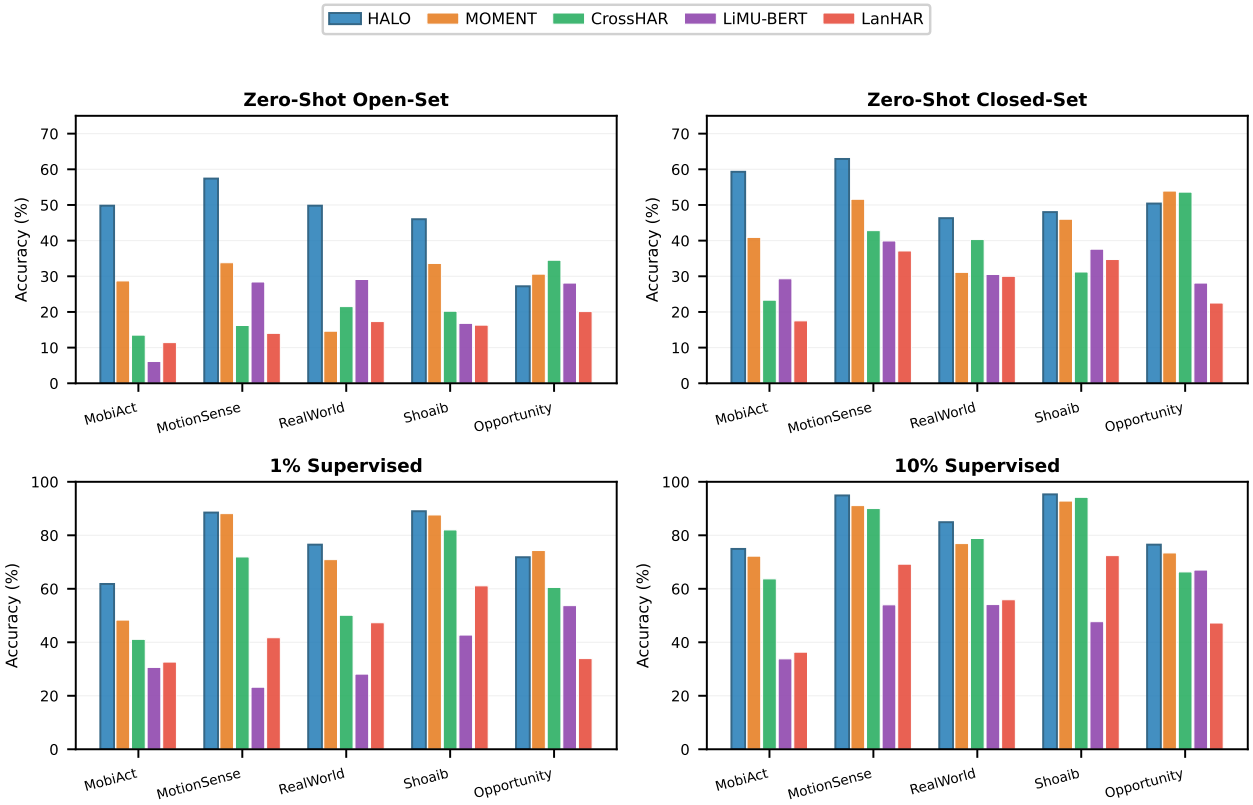

Figure 4: Per-dataset accuracy (%) across all four evaluation settings for the five main test datasets. HALO (blue, leftmost bar) leads on most dataset–metric combinations. Opportunity is the only dataset where MOMENT leads in zero-shot closed-set (53.9% vs. 50.4%), aided by its 6144-dim SVM-RBF classifier on this small, well-separated label set.

label coverage, this sensor distribution shift causes all models to collapse in zero-shot. VTT-ConIoT is an industrial/construction dataset where only 8 of 16 activities (50%) have training equivalents; the remaining activities (e.g., kneeling work, roll painting, climbing ladder) are completely novel.

	HARTH				VTT-ConIoT			
	ZS-O	ZS-C	1%	10%	ZS-O	ZS-C	1%	10%
HALO	**1.4**	0.4	**69.2**	**79.1**	1.9	6.3	10.6	26.1
MOMENT	0.4	1.9	66.7	65.4	1.6	5.2	**18.4**	**34.8**
CrossHAR	0.3	0.9	42.4	71.9	0.7	5.0	10.6	30.9
LiMU-BERT	0.0	**20.2**	54.4	19.1	3.4	**7.1**	4.3	15.0
LanHAR	0.9	1.1	3.4	70.0	**8.3**	6.9	7.7	7.7
LLaSA†	11.9	12.5	—	—	—	6.9	—	—

Table 6: Severe OOD accuracy (%). ZS-O/ZS-C: zero-shot open/closed-set. All models collapse on HARTH zero-shot (sensor distribution shift); VTT-ConIoT is capped at ∼50% by novel activities. †LLaSA: zero-shot only, subsampled.

On HARTH, all models achieve near-zero zero-shot accuracy (<2% for HALO, MOMENT, CrossHAR), confirming genuine sensor distribution shift rather than a model-specific failure. LiMU-BERT's closed-set anomaly (20.2%) likely reflects its GRU classifier exploiting a frequency bias in the masked logits rather than genuine activity discrimination. With supervised data, HALO leads at both 1% (69.2% vs. MOMENT's 66.7%) and 10% (79.1% vs. CrossHAR's 71.9%). This suggests HALO's encoder representations are flexible enough to adapt to novel sensor configurations given even minimal labeled data.

On VTT-ConIoT, all models score near random on zero-shot (<8%). With supervised data, MOMENT leads (34.8% at 10%), benefiting from its general time-series pretraining across 13 diverse domains. HALO's HAR-specific architecture provides strong in-domain transfer but less benefit when the target domain is fundamentally different from training data.

6.2.4 Embedding Space Analysis. Figure 5 visualizes the learned embedding space at the end of Stage 2 training using UMAP [18]. Each activity group is shown with its IMU centroid (circle) and text centroid (diamond), connected by a line. Short connecting lines indicate tight cross-modal alignment; spatially proximate groups reflect semantic similarity. The visualization shows that HALO successfully pulls sensor and language representations into a shared geometry: semantically related groups (e.g., *running*, *stairs*, *walking*) cluster together, while distinct activities (e.g., *cycling*, *falling*) are well-separated. The embedded statistics report nearest-neighbor accuracy of 76.7%, mean positive similarity of 0.857, and similarity gap (positive minus negative) of 0.377, confirming that the contrastive objective produces a well-structured, discriminative space at convergence.

6.3 Ablation Study

HALO's architecture supports dynamic handling of variable sampling rates and per-channel semantic descriptions. To quantify this contribution, we compare HALO evaluated at native 50 Hz with rich manifest-derived channel descriptions against HALO evaluated on the same 20 Hz resampled data as baselines with generic channel descriptions ("Accelerometer X-axis"). This ablation was measured on the original three main test datasets (all 50 Hz). Figure 6 reports the accuracy deltas, with per-dataset breakdown below.

The effects are large but dataset-dependent. MotionSense and MobiAct show massive zero-shot gains (+24.5 and +19.7 open-set respectively): prepending the dataset description to channel text matches the training format and lets the model leverage sensor placement and device context for disambiguation. RealWorld, however, drops significantly in zero-shot (–9.9 open-set, –11.9 closed-set): RealWorld collects data from multiple heterogeneous body placements with a single dataset-level description that over-commits the model to one sensor context. This is an honest negative result that highlights a current limitation of dataset-level descriptions for heterogeneous data. Supervised performance is much less affected by channel conditioning (deltas of –0.5 to +12.2 at 1%, and within 3 points at 10%), consistent with the expectation that labeled data enables the model to learn dataset-specific patterns directly.

Net over three datasets, the zero-shot open-set average improves by +11.4 percentage points despite RealWorld's regression, demonstrating that HALO's channel-text conditioning is a genuinely useful architectural capability. The RealWorld result points toward per-sample or per-subject metadata conditioning as a direction for future work.

6.4 Patch Size Sensitivity

HALO uses a fixed 1.0 s patch size for all reported results—no per-dataset sweep or test-time tuning. Figure 7 evaluates zero-shot closed-set accuracy across candidate patch sizes. The original three datasets were measured at 20 Hz; the expanded benchmarks (Shoaib, Opportunity) are measured at native rates (50 Hz and 30 Hz respectively).

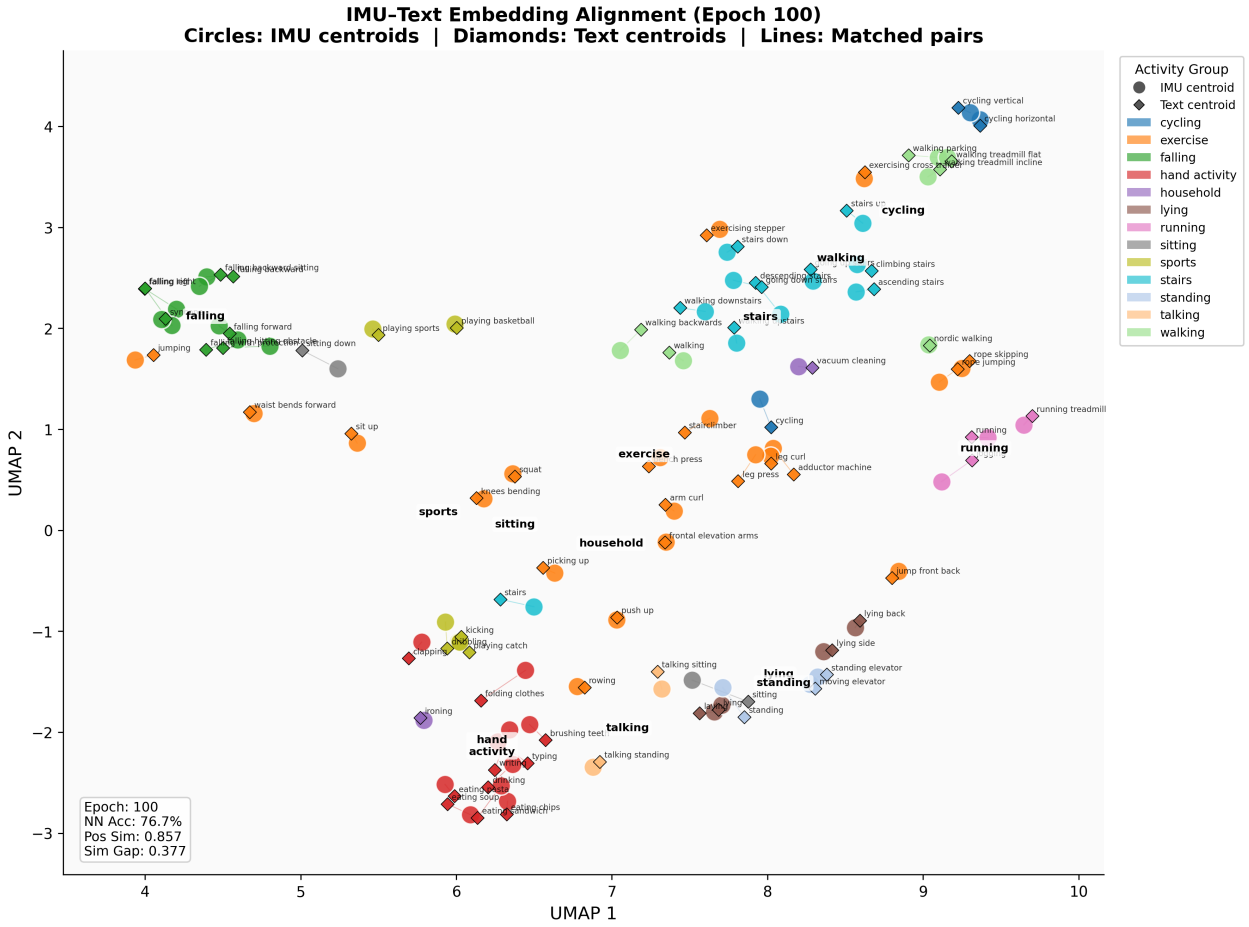

Figure 5: UMAP projection of learned IMU and text embeddings at epoch 100. Circles show per-activity IMU centroids; diamonds show frozen SBERT text centroids; lines connect matched pairs. Colors denote semantic activity groups. Short lines and co-located markers indicate tight cross-modal alignment. Stats (inset): nearest-neighbor accuracy 76.7%, positive cosine similarity 0.857, similarity gap 0.377.

Smaller patches (1.0–1.25 s) consistently perform best. The fixed 1.0 s choice is within 1.1 percentage points of the per-dataset optimum for MotionSense, RealWorld, and MobiAct. On the expanded benchmarks at native rates, 1.0 s is within 0.9 pp of the optimum on Shoaib and within 1.3 pp on Opportunity. Opportunity's slight preference for 1.5 s is physically sensible: at 30 Hz, a 1.5 s patch contains 45 samples, close to the 50 samples that 50 Hz datasets receive with a 1.0 s patch. HARTH's near-zero zero-shot performance makes its patch sensitivity uninformative.

7 DISCUSSION

Language alignment enables open-vocabulary zero-shot recognition. HALO achieves 46.0% average open-set accuracy across five main test datasets, ahead of MOMENT (28.3%) despite 10× fewer trainable parameters (∼35M vs. 341.2M). Classifier-based models can only output training labels, while text-aligned models score any description at test time. LLaSA's poor closed-set performance (17.4%) confirms that scaling to a 7B LLM does not compensate for the limitations of generative classification on IMU data.

Channel-text conditioning reveals a metadata granularity limitation. Native-rate evaluation with rich

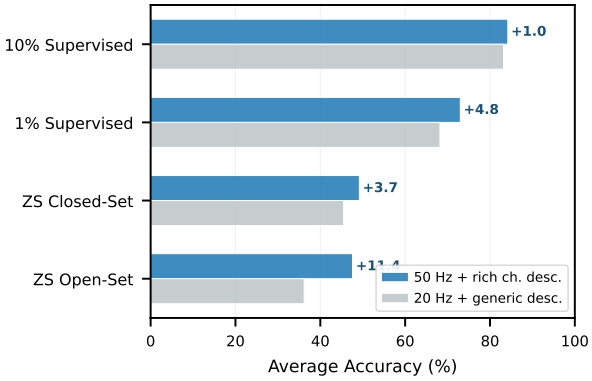

Figure 6: Native rate ablation: average accuracy (%) across 3 main test datasets. Blue bars show HALO at native 50 Hz with rich channel descriptions; gray bars show 20 Hz resampled with generic descriptions. Annotations show the accuracy gain from native-rate evaluation.

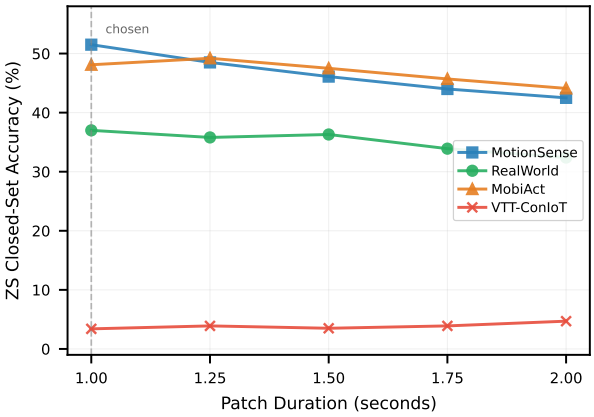

Figure 7: Patch size sensitivity: zero-shot closed-set accuracy (%) at each candidate patch duration. The dashed vertical line marks the fixed 1.0 s choice. Original datasets measured at 20 Hz; Shoaib/Opportunity at native rates.

channel descriptions improves average zero-shot accuracy by 11.4 points over 20 Hz with generic descriptions. MotionSense and MobiAct gain substantially (+24.5 and +19.7 points), but RealWorld drops 9.9 points because its heterogeneous body positions are poorly captured by a single dataset-level description, motivating per-sample metadata conditioning as future work.

Severe distribution shift remains an open challenge. On HARTH and VTT-ConIoT, all models collapse in zero-shot (<2% and <8%). HALO adapts effectively on HARTH with supervision (79.1% at 10%), while MOMENT leads on VTT-ConIoT (34.8%), suggesting general time-series pretraining provides broader priors for truly novel domains. Key limitations include dataset-level channel descriptions, a 384-dim embedding bottleneck, and an 87-label training vocabulary. Future work targets per-sample sensor conditioning, multi-prototype label pooling, and unsupervised domain adaptation.

8 CONCLUSION

We presented HALO, an IMU foundation model trained across 10 heterogeneous datasets via self-supervised pretraining and CLIP-style IMU-to-text alignment. Seconds-based patch tokenization, channel-independent feature extraction, and perceiver-style attention enable handling of variable sampling rates, channel counts, and label vocabularies without dataset-specific modifications. HALO leads five baselines on all 8 aggregate metrics with ∼35M trainable parameters, and adapts effectively to severe distribution-shift datasets with minimal supervision, supporting language as a practical label interface for open-set wearable sensing.

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